

Miguel L. Martins

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Research Interests

Model-based learning, Representation Learning Theory, Robust/Adversarial Learning, Self-supervised Learning, Multimodal Learning and Alignment, Signal Processing, Useful theory

Education

CentraleSupélec, Paris-Saclay

Paris, France

Visiting Ph.D. Student

2025–2026

- Bridge no free-lunch theorems from inverse problems to self-supervised identifiability theory
- Multi-modal alignment of VLMs and its impact in downstream performance
- Understanding the model capacity of pre-trained embedders: accessing model capacity without access to labels
- Help mentor CentraleSupélec and ILLS students in projects related to certified robustness and uncertainty quantification

University of Porto

Porto, Portugal

Ph.D. in Computer Science

2021–2026

- Thesis: *Geometrical and Statistical Priors for Deep Learning Models with Biomedical Applications*
- Topics: Model-based Learning, Multifractal formalism, Hidden Markov Models, Representation Theory, Biomedical Modalities

University of Porto

Porto, Portugal

M.Sc. in Computer Science - Data Science Specialization

2017–2019

- Thesis: *Condensed Graphs: Towards a General Approach for Faster Subgraph Census*

University of Coimbra

Coimbra, Portugal

B.Sc. in Informatics Engineering

2013–2016

Highlight Publications

2025: M. L. Martins, M. T. Coimbra and F. Renna *Monofractal and Multifractal Recalibration of Fully Convolutional Networks for Medical Image Segmentation*. Under review for **SIAM Journal on Imaging Sciences**. It is a compendium of the works developed within the scope of my second “big” project of my PhD. Preprint available on [arXiv](#).

2023: M. L. Martins, M. T. Coimbra and F. Renna *Markov-Based Neural Networks for Heart Sound Segmentation: Using Domain Knowledge in a Principled Way*. **IEEE Journal of Biomedical and Health Informatics**.

The complete list of my international peer-reviewed publications can be consulted on my [Google Scholar](#).

Tutorials and Scientific Communication

ELLIS Robust Machine Learning Group

University of Tübingen

Identifiability is not a free lunch for self-supervised learning

2026

French Alt. Energies and Atomic Energy Comm. (CEA), List

Université Paris-Saclay

Identifiability results in self-supervised learning from a robustness perspective

2025

Invited Talk + Tutorial Session

Tutorial on Deep learning techniques for digital cardiac auscultation analysis

- **2025:** IEEE International Conference on Body Sensor Networks
- **2025:** University of Coimbra, 20th National Gathering of Biomedical Engineering Students (ENEED)
- **2024:** EUGLOH BIP Advanced Processing of Biomedical Signals and Images, University of Porto

Data Science Portugal Talks

Porto Tech Hub

From Graph Theory to Modern Network Science

2020

Teaching Experience

Discrete Mathematics

University of Porto

Teaching Assistant

2022/2023

Professional Experience

Data Scientist

Critical TechWorks & BMW

Natural Language Processing and Understanding

2019-2021

- Research and develop state-of-the-art Transformer architectures for sentence2sentence tasks
- Fine-tuning on multi-lingual, domain-specific vocabulary
- Ranking and Search Optimization for Information Retrieval

Scholarships and Awards

2021: Best Oral Presentation Award (*RECPAD 2021*)

2021: Best In Person Poster Presentation Award (*Computing in Cardiology*)

2021: FCT PhD. Scholarship Reference 2021.06503.BD

Short bio

My name is Miguel, and I am about to defend my PhD in Computer Science at the University of Porto. If I had to describe my trajectory so far, it would be something like a “breadth-first search.” I started with a fairly classical computer science background—my MSc focused on algorithms for subgraph census, trying to make an NP-complete problem a bit more tractable in real-world networks. After that, I moved to industry and worked as a data scientist in a BMW joint venture, right at the moment when Transformers were becoming mainstream. I worked on semantic equivalence between documents using ensembles of distilled Transformer models, along with tasks like topic modeling and summarization. It was fast-paced, especially during COVID, and while I learned a lot, I often found myself moving on from systems without fully understanding why they worked—or didn't. That discomfort is what pushed me back to academia.

For my PhD, I joined a biomedical research group, though my focus was always more on the underlying algorithms than on the application domain itself. I was particularly drawn by the

information-theoretic perspective of my advisor. Like many biomedical projects, progress depended heavily on data that did not always arrive, but I was given the freedom to explore my own ideas. This led me into somewhat unconventional directions, such as multifractal analysis. Over time, however, I became less interested in purely chasing performance and more concerned with understanding the principles behind these models.

Towards the end of my PhD, I had the opportunity to pursue a more self-directed line of work, which led me to CentraleSupélec. There, I started exploring connections between inverse problems and self-supervised learning, particularly through the lens of identifiability and no-free-lunch theorems. What began as an attempt to explain the robustness of self-supervised methods evolved into a more theoretical investigation, which has proven quite fruitful and is currently being prepared for submission.

During my time in Paris, I became involved in several collaborations. One of my main projects studies whether self-supervised objectives can be used to estimate the capacity of pre-trained models—essentially predicting downstream performance without labels. We have promising empirical results and are now working on the theoretical grounding. I am also collaborating on multimodal learning, particularly modality alignment in VLMs. We already submitted some preliminary work on a more applied basis to medical conference but me and my colleague are thinking about extending the results from a more fundamental perspective to NeurIPS.

More recently, I've been involved in projects on domain alignment in CLIP-based models and on foundational models for tabular data imputation in collaboration with MIT, Harvard, and Google. Across all of these, the common thread is the same: understanding representations—not just improving them, but making sense of what they encode, how they generalize, and how they can be trusted.

On top of that I still have a great relationship with my research group, and often help my (soon to be) former advisor with writing research projects and coordinating students.

I intentionally leverage my experience across multiple problem domains as a strength at this stage of my research career. That being said, my intention is to pivot more and more to the more fundamental and deliberate side of machine learning research.